

UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE
Department of Computer Science – Software Engineering
ADVANCED WEB DEVELOPMENT

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NRC: 30716

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Accounting System – Product Backlog & Features Document

Product Backlog Decomposition into Features by Module

1. Features Summary Table

ID	Feature Name	Primary Module	Sprint(s)	Lead(s)	Effort
FT-01	User Identity & Secure Access	Mod 1	1	D.R / E.Q	7 hrs
FT-02	Workspace & Project Isolation	Mod 1	1	A.S	7 hrs
FT-03	SRI Service Handshake	Mod 2	2-3	E.Q	7 hrs

FT-04	Automated Invoice Retrieval	Mod 3	4	E.Q	4 hrs
FT-05	Fiscal Data Timeline Filtering	Mod 3	5	D.R	4 hrs
FT-06	Local Import & Ownership Integrity	Mod 3	5-6	A.S / D.R	6 hrs
FT-07	ATS XLSTM Generation Engine	Mod 4	7-8	E.Q / A.S	12 hrs
FT-08	XLSTM Structural Diagnostics	Mod 5	9	E.Q / A.S	7 hrs
FT-09	Official XML Encoding (DIMM SRI)	Mod 6	10-11	E.Q / A.S / D.R	14 hrs
FT-10	Workflow Traceability Dashboard	Mod 7	12	E.Q / A.S	14 hrs

2. Feature Details & Specification

FT-01 – User Identity & Secure Access

Field	Detail
Description	Handles taxpayer registration, login, and data protection via encrypted sessions.
Objective	Secure the system and ensure data belongs to the authenticated RUC.
Module	Module 1 – Configuration & Setup
Associated RFs	RF-01
Responsible	David Rodriguez / Esteban Quiroga
Business Rules	RN-02 (Ownership validation)

Associated User Stories

ID	User Story	Actor	Priority	Status

US-01	As a user, I want to register and log in to protect my information.	Taxpayer	High	Prioritized
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Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-01-01	Implement registration and login interface.	US-01	David Rodriguez	Sprint 1	5
T-01-03	Apply data protection for saved documents.	US-01	Esteban Quiroga	Sprint 1	2

FT-02 – Workspace & Project Isolation

Field	Detail
Description	Creation of isolated local directories based on specific taxpayer RUCs and periods.
Objective	Prevent cross-contamination between different fiscal periods.

Module	Module 1 – Configuration & Setup
Associated RFs	RF-02, RF-03
Responsible	Angel Sabando
Business Rules	RN-01 (Tax periods)

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-01-04	Implement directory creation and file path mapping for workspaces.	US-03	Angel Sabando	Sprint 1	7

FT-03 – SRI Service Handshake

Field	Detail
Description	Secure connection and authentication with the SRI online portal.

Objective	Establish a functional link to the external government web page.
Module	Module 2 – SRI Connectivity
Associated RFs	RF-02
Responsible	Esteban Quiroga

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-02-01	Connect the system with the SRI portal.	US-02	Esteban Quiroga	Sprint 2	7

FT-04 – Automated Invoice Retrieval

Field	Detail
Description	Automated download of electronic invoices in XML/PDF format directly from SRI.

Objective	Eliminate manual document gathering.
Module	Module 3 – Data Acquisition
Associated RFs	RF-03
Responsible	Esteban Quiroga

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-03-01	Automate the download of invoices from the portal.	US-04	Esteban Quiroga	Sprint 4	4

FT-05 – Fiscal Data Timeline Filtering

Field	Detail
Description	Filter logic to ensure invoices match the monthly or semiannual reporting range.

Objective	Exclude invoices outside the selected fiscal period.
Module	Module 3 – Data Acquisition
Associated RFs	RF-04
Responsible	David Rodriguez
Business Rules	RN-01

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-03-03	Develop temporal filter based on XML "fechaEmision" tags.	US-04	David Rodriguez	Sprint 5	4

FT-06 – Local Import & Ownership Integrity

Field	Detail
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Description	Manual import of external XMLs with RUC validation.
Objective	Allow inclusion of local files while maintaining identity integrity.
Module	Module 3 – Data Acquisition
Associated RFs	RF-04
Responsible	Angel Sabando / David Rodriguez
Business Rules	RN-02

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-03-02	Implement manual upload and RUC validation.	US-05	Angel Sabando	Sprint 6	4

Field	Detail
Description	Mapping of invoice data into a standardized XLSM spreadsheet.
Objective	Transform raw XML data into an editable accounting format.
Module	Module 4 – Core Transcription
Associated RFs	RF-05
Responsible	Esteban Quiroga / Angel Sabando

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-04-01	Implement Excel transcription logic.	US-06	Esteban Quiroga	Sprint 7	6

T-04-02	Map XML tags to XLSM headers.	US-06	Angel Sabando	Sprint 8	6
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FT-08 – XLSM Diagnostic

Field	Detail
Description	Let the user modify the generated XLSM for errors and then validate it before XML export.
Objective	Reduce final export failures.
Module	Module 5 – Validation
Associated RFs	RF-06
Responsible	Esteban Quiroga / Angel Sabando

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs

T-05-01	Develop XLSM file validation.	US-07	Esteban Quiroga	Sprint 9	5
T-05-02	Design log system for error notifications.	US-07	Angel Sabando	Sprint 9	2

FT-09 – Official XML Encoding (DIMM SRI)

Field	Detail
Description	Conversion of XLSM data into SRI-compliant XML (DIMM format).
Objective	Generate the official legal file for submission.
Module	Module 6 – XML Generation
Associated RFs	RF-08
Responsible	E. Quiroga / A. Sabando / D. Rodriguez

NFRs	NFR-08, NFR-10
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Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-06-01	Develop XML creation from validated XLSM.	US-08	Esteban Quiroga	Sprint 10	5
T-06-02	Validate against SRI DIMM schema.	US-08	Angel Sabando	Sprint 11	5
T-06-03	Implement secure XML file download.	US-08	David Rodriguez	Sprint 11	4

FT-10 – Workflow Traceability Dashboard

Field	Detail
Description	Visual status tracking of the entire ATS generation process.

Objective	Provide real-time monitoring of the workflow stages.
Module	Module 7 – Traceability
Associated RFs	RF-09
Responsible	Esteban Quiroga / Angel Sabando

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-07-01	Create status widget (completed/pending/blocked).	US-09	Esteban Quiroga	Sprint 12	7
T-07-02	Connect widget to all modules for automatic updates.	US-09	Angel Sabando	Sprint 12	7

3. Effort Distribution Summary

Team Member	Assigned Tasks	Estimated Total
Esteban Quiroga	T-01-03, T-02-01, T-03-01, T-04-01, T-05-01, T-06-01, T-07-01	36 hrs
Angel Sabando	T-01-04, T-03-02, T-04-02, T-05-02, T-06-02, T-07-02	34 hrs
David Rodriguez	T-01-01, T-01-02, T-03-03, T-06-03	30 hrs
Total Project	Full Backlog Implementation	100 hrs